THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 8 · Term 1

Student edition

Outcomes: MA-8.1 rational and irrational numbers, significant figures, laws of exponents; MA-8.7 discount, markup, insurance, partnership, inheritance shares and currency conversion.

Lesson notes

- **Rational and irrational (Weeks 1–4):** rational numbers can be written as p/q (terminating or recurring decimals, e.g. 0.333... = 1/3); irrational numbers cannot (√2, π). Estimate square roots; place on a number line.
- **Exponents and significant figures (Weeks 5–8):** $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$. Round to significant figures (3,476 to 2 s.f. = 3,500; 0.004 52 to 2 s.f. = 0.0045). Link to scientific notation.
- **MA-8.7 Commercial maths (Weeks 9–12):** discount % on the marked price; markup on cost price; insurance premium as a percentage of the sum insured; partnership profit shared in the ratio of capital; currency conversion at a given rate (use the day's rate from the newspaper). Inheritance: one basic case from Surah An-Nisa (4:11) — a son's share is twice a daughter's.

Workbook pages

Page	Task
1	Sort 16 numbers into rational and irrational.
2–3	Exponent laws and significant figures.
4–5	Shop discount and markup problems; partnership and currency problems.

Question bank

#	Outcome	Question	Your answer
1	MA-8.1	Which is irrational?	0.5 / 1/3 / √2 / √9
2	MA-8.1	0.333... (recurring) is a rational number. True or false.	True / False
3	MA-8.1	$2^3 \times 2^4 = ?$	2^{12} / 2^7 / 4^7 / 2^1
4	MA-8.1	$5^0 = ?$	0 / 5 / 1 / undefined
5	MA-8.1	$3^{-2} = ?$	−9 / −6 / 1/9 / 9
6	MA-8.1	3,476 to 2 significant figures?	3,400 / 3,500 / 35 / 3,480
7	MA-8.7	Marked price Rs 2,000 with 15% discount. Sale price?	Rs 1,850 / Rs 1,700 / Rs 300 / Rs 2,300

#	Outcome	Question	Your answer
8	MA-8.7	Cost Rs 800, markup 25%. Selling price?	Rs 825 / Rs 900 / Rs 1,000 / Rs 1,200
9	MA-8.7	A and B invest Rs 30,000 and Rs 20,000. Profit Rs 10,000. A's share?	Rs 5,000 / Rs 6,000 / Rs 4,000 / Rs 7,500
10	MA-8.7	At an example rate of 1 US dollar = Rs 280, how many rupees is 50 dollars?	Rs 5,600 / Rs 14,000 / Rs 1,400 / Rs 330